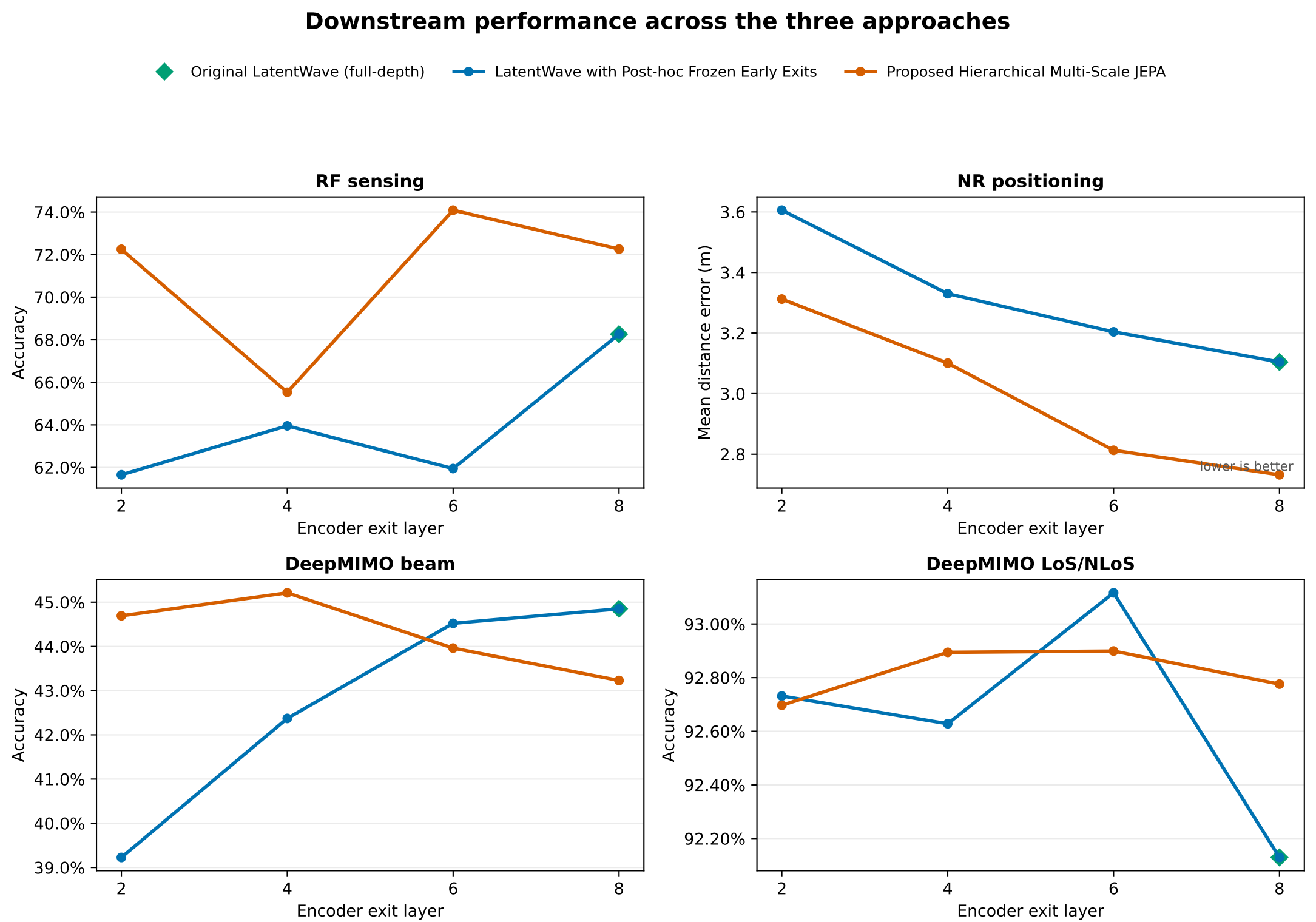
Downstream performance across the three approaches
Original LatentWave (full-depth)
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Accuracy
74.0%
72.0%
70.0%
68.0%
66.0%
64.0%
62.0%
2
4
6
8
Encoder exit layer
NR positioning
Mean distance error (m)
3.6
3.4
3.2
3.0
2.8
lower is better
Encoder exit layer
DeepMIMO beam
Accuracy
45.0%
44.0%
43.0%
42.0%
41.0%
40.0%
39.0%
Encoder exit layer
DeepMIMO LoS/NLoS
Accuracy
93.00%
92.80%
92.60%
92.40%
92.20%
Encoder exit layer
Diamond: Original LatentWave at full-depth L8. Circles: frozen-head exits. All rows use the shared 80/10/10 split and frozen context encoders.